

		maximum velocity $= \omega x_0 = \sqrt{8} \times 0.5 = 1.41 \text{ ms}^{-1}$
17	D	$I = \frac{P}{4\pi r^2}$ At 3.0 m, $P = 0.18 \times 4\pi \times 9 = 6.48\pi$ Power is now tripled, new $P = 19.44\pi$ At 4.5 m, $I = \frac{P}{4\pi r^2} = \frac{19.44\pi}{4\pi 4.5^2} = 0.24 \text{ W m}^{-2}$
18	C	For stationary wave, a displacement node corresponds to a pressure antinode hence the